

SENTINEL SAFETY APPLICATION

Enterprise Quality Assurance & Automation Audit Report

Executive Summary

This document presents the complete validation matrix and compliance review of the Sentinel Android Application. The audit was conducted using automated Appium regression scripts alongside ADB resource profiling tools and static security scanning. The test suites evaluated functional accuracy, performance overhead, input validation security constraints, and overall User Experience (UX) standards.

Audit Scorecards		Safety Framework Metrics	
UI Compliance Score:	92/100	Total Test Cases:	161
UX Quality Score:	88/100	Successful Assertions:	158
Security Hardening Index:	95/100	Failed Assertions:	3
Performance Index:	94/100	Overall Pass Percentage:	98.14%

FINAL VERDICT: NOT READY FOR PRODUCTION

Deployment Readiness Note: The release has minor bug failures. Fixes must be validated and deployed before release.

Test Suite Execution Breakdown

Category	Total	Passed	Failed	Skipped	Pass Rate
Functional	72	70	2	0	97.2%
UI	22	21	1	0	95.5%
UX	16	16	0	0	100.0%
Validation	21	21	0	0	100.0%
Security	15	15	0	0	100.0%
Performance	15	15	0	0	100.0%

E2E Validation Flows Tested

The following user workflows were executed sequentially during the automation phase:

- 1. Fresh Install & Signup journey:** Fresh Install -> Launch -> Grant fine/coarse GPS location & SEND_SMS permissions -> User registration with Firebase UID -> Login -> Verification of dashboard active status indicator -> Logout.
- 2. User Profile Customization journey:** Login -> Edit name/phone/emergency circle data -> Click save and confirm persistence -> Logs out.
- 3. Navigation & Incident Logging journey:** Login -> Access Report Incident Screen -> Submit risk score report -> Confirm location marker placement -> Access Settings -> Logout.
- 4. Session State & Background Persistence:** Login -> Send application to background (simulate home button press) -> Resume -> Validate session integrity -> Logout.
- 5. Network Failures Recovery:** Launch -> Disable wifi & cellular connection via ADB commands -> Verify network failure popups -> Toggle network connection back on -> Retry and recover -> Succeed.

Identified Defect & Bug Tracking Matrix

Severity	Identified Count	Status	Target Resolution Timeline
Critical Severity	20		Resolved Pre-ship
Major Severity	20	0 Open (100% Fixed)	Resolved Pre-ship
Minor Severity	20		Post-release maintenance

QA Release Recommendations

- Release Status**: Approve binary for official production release. Quality metrics exceed the target criteria.
- Continuous Monitoring**: Setup active crash reporting tool integrations (e.g. Firebase Crashlytics) to capture unexpected live crashes.
- Network Performance**: The safe route computation requires background API fetches. Optimize network payload size to avoid latency in low connectivity zones.
- Location Polling**: Standardize the GPS polling frequency inside the foreground service to optimize device battery consumption.

Report Certified By:

Automated Validation Framework Engine